



## Solve with us 9 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- Scalar-valued multivariable functions:Scalar-valued multivariable functions: A scalar-valued multivariable function is a function  $f: D \rightarrow \mathbb{R}$  where  $D$  is a domain in  $\mathbb{R}^n$  where  $n > 1$ .
- Vector-valued multivariable functions:Vector-valued multivariable functions: A vector-valued multivariable function is a function  $f: D \rightarrow \mathbb{R}^m$  where  $D$  is a domain in  $\mathbb{R}^n$  where  $n, m > 1$ .

1 point

Which of the following functions are scalar-valued multivariable functions?

☐

$$f(x, y) = xy + \sin(xy)$$

☐

$$f(x, y, z) = \ln(x + y + z)$$

☐

$$f(x, y, z) = (x^2, z, y + xz)$$

☐

$$f(x, y) = (xy, 0)$$

☐

$$f(x, y) = xy$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f(x, y) = xy + \sin(xy)$$

$$f(x, y, z) = \ln(x + y + z)$$

$$f(x, y) = xy$$

1 point

Which of the following functions are vector-valued multivariable functions?

☐

$$f(x, y) = xy + \sin(xy)$$

☐

$$f(x, y, z) = \ln(x + y + z)$$

☐

$$f(x, y, z) = (x^2, z, y + xz)$$

☐

$$f(x, y) = (xy, 0)$$

☐

$$f(x, y) = xy$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f(x, y, z) = (x^2, z, y + xz)$$

$$f(x, y) = (xy, 0)$$



Key Points: 2Key Points: 2

- Composition of functions:Composition of functions: Let  $D \subseteq \mathbb{R}^n$   $D \subseteq \mathbb{R}^n$  and  $f: D \rightarrow \mathbb{R}^m$   $f: D \rightarrow \mathbb{R}^m$  be a multivariable function. Let  $g: E \rightarrow \mathbb{R}^p$   $g: E \rightarrow \mathbb{R}^p$  be a function on  $E$  where  $\text{Range}(f) \subseteq E \subseteq \mathbb{R}^m$   $\text{Range}(f) \subseteq E \subseteq \mathbb{R}^m$ .

We define a multivariable function  $g \circ f: D \rightarrow \mathbb{R}^p$   $g \circ f: D \rightarrow \mathbb{R}^p$  called the composition of  $f$  and  $g$  defined as  $g \circ f(x) = g(f(x))$   $g \circ f(x) = g(f(x))$ , for all  $x \in D$   $x \in D$ .

1 point

If two functions  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$   $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  and  $g: \mathbb{R} \rightarrow \mathbb{R}$   $g: \mathbb{R} \rightarrow \mathbb{R}$  are defined as  $f(x, y) = (x - y)$   $f(x, y) = (x - y)$  and  $g(x) = \sin x$   $g(x) = \sin x$ , then which of the following functions represents  $g \circ f$   $g \circ f$ ?

☐

$\sin x - \sin y$   $\sin x - \sin y$ .

☐

$\sin(x - y)$   $\sin(x - y)$ .

☐

$x \sin x - y \sin y$   $x \sin x - y \sin y$ .

☐

$\sin x y \sin xy$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\sin(x - y)$   $\sin(x - y)$ .

1 point

If two functions  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$   $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  and  $g: \mathbb{R} \rightarrow \mathbb{R}^2$   $g: \mathbb{R} \rightarrow \mathbb{R}^2$  are defined as  $f(x, y) = xy$   $f(x, y) = xy$  and  $g(t) = (e^t, e^{-t})$   $g(t) = (e^t, e^{-t})$ , then which of the following functions represents  $g \circ f$   $g \circ f$ ?

☐

$g \circ f(x, y) = e^{xy}$   $g \circ f(x, y) = e^{xy}$ .

☐

$g \circ f(x, y) = 1$   $g \circ f(x, y) = 1$ .

☐

$g \circ f(x, y) = (e^{xy}, e^{-xy})$   $g \circ f(x, y) = (e^{xy}, e^{-xy})$ .

☐

$g \circ f(x, y) = (e^x, e^{-x})$   $g \circ f(x, y) = (e^x, e^{-x})$ .

☐

$g \circ f(x, y) = (1, 1)$   $g \circ f(x, y) = (1, 1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$g \circ f(x, y) = (e^{xy}, e^{-xy})$   $g \circ f(x, y) = (e^{xy}, e^{-xy})$ .

Key Points: 3Key Points: 3

- Partial derivatives: Partial derivatives: Let  $f(x_1, x_2, \dots, x_n)$  be a function defined on a domain  $D$  in  $\mathbb{R}^n$ . The partial derivative of  $f$  with respect to  $x_i$  is the function denoted by  $f_{x_i}(x)$  or  $\frac{\partial f}{\partial x_i}(x)$  and defined as

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \rightarrow 0} \frac{f(x + he_i) - f(x)}{h}$$

where  $e_i$  is the unit vector  $(0, 0, \dots, 0, 1, 0, \dots, 0)$  (at  $i$ -th coordinate there is 1, and 0 at all the other coordinates).

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f((x_1, x_2, \dots, x_n) + h(0, 0, \dots, 0, 1, 0, \dots, 0)) - f(x_1, x_2, \dots, x_n)}{h}$$

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

- To calculate the partial derivative with respect to  $x_i$ , think of  $f$  only as a function of  $x_i$  while treating all other variables as constants. Then calculate it as the derivative of a function of one variable.  
1 point  
Consider the function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  such that  $f(x, y) = x^2 + y^2$ . Which of the following function will represent  $f_x$  or  $\frac{\partial f}{\partial x}$ ?

☐

$2x + 2y$ .

☐

$2x$ .

☐

$2x + y^2$ .

☐

$x + y^2$ .

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

Treat  $y$  as constant.

**Accepted Answers:**

$2x$ .

1 point

Consider the function  $f: D \rightarrow \mathbb{R}$  where  $D \subset \mathbb{R}^2$ , defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$(0, 0)$

What will be the partial derivative of  $f$  with respect to  $x$  at the point  $(0, 0)$ ?

☐

11.

☐

00.

☐

22

☐ Cannot be determined.

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

00.

Key Points: 4Key Points: 4

- Directional derivative: Directional derivative: The rate of change of  $f(x_1, x_2, \dots, x_n)$  in the direction of the unit vector  $u = (u_1, u_2, \dots, u_n)$  is called the directional derivative and is denoted by  $D_u f(x_1, x_2, \dots, x_n)$ . The definition of the directional derivative is:

$$D_u f(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1 + u_1 h, x_2 + u_2 h, \dots, x_n + u_n h) - f(x_1, x_2, \dots, x_n)}{h}$$

- To calculate the directional derivative of a function  $f$  in a particular direction, we have to first find the unit vector  $u$  in that direction and then compute  $D_u f$ .
- The partial derivatives are special cases of the directional derivative where  $u = e_i$  where  $e_i$  is  $i$ -th vector of the standard ordered basis of  $R^n$ .

1 point

Consider the function  $f: R^2 \rightarrow R$ , such that  $f(x, y) = x + y$ . Which of the following options will denote the directional derivative of  $f$  in the direction of the vector  $v = (1, 1)$ ?

[Hint: The given vector  $v$  is not a unit vector. Convert  $v$  to a unit vector  $u = \frac{v}{\|v\|}$ .]

☐ 2

☐

$x + y$

☐

$\sqrt{2}$

☐

☐

$\sqrt{2}(x + y)$

☐

$(x + y)$

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

$\sqrt{2}$

☐

## Key Points: 5Key Points: 5

- Limit of a scalar valued multivariable function: Limit of a scalar valued multivariable function: If there exists some real number  $L$  such that  $f(x_1, x_2, \dots, x_n)$  approaches  $L$  as  $(x_1, x_2, \dots, x_n)$  approaches  $(a_1, a_2, \dots, a_n)$ , then we write
 
$$\lim_{(x_1, x_2, \dots, x_n) \rightarrow (a_1, a_2, \dots, a_n)} f(x_1, x_2, \dots, x_n) = L$$
- For functions of one variable  $f(x)$ , there are only two ways in which  $x$  can approach a point  $a$ , from left or from right. But for multivariable functions,  $(a_1, a_2, \dots, a_n)$  can be approached along any curve.
- If  $C$  is a curve passing through  $(a_1, a_2, \dots, a_n)$ , the limit of  $f$  at  $(a_1, a_2, \dots, a_n)$  along the curve  $C$  exists and equals  $L$  if  $f(x_1, x_2, \dots, x_n)$  approaches  $L$  as  $(x_1, x_2, \dots, x_n)$  approaches  $(a_1, a_2, \dots, a_n)$  where  $(x_1, x_2, \dots, x_n)$  belongs to  $C$ .
- The limit of the function at  $(a_1, a_2, \dots, a_n)$  along the curve  $C$  equals  $L$  for each curve  $C$  passing through  $(a_1, a_2, \dots, a_n)$  precisely when the limit of  $f$  at  $(a_1, a_2, \dots, a_n)$  equals  $L$ .
- To show that the limit of  $f$  at  $(a_1, a_2, \dots, a_n)$  does not exist, one should either produce a curve  $C$  so that the limit of  $f$  at  $(a_1, a_2, \dots, a_n)$  along  $C$  does not exist or produce two curves so that the limits of  $f$  at  $(a_1, a_2, \dots, a_n)$  along both curves are not equal.
- Continuity of a function: Continuity of a function: If limit of a function  $f: D \rightarrow \mathbb{R}$ , where  $D \subseteq \mathbb{R}^n$ , exists at a point  $(a_1, a_2, \dots, a_n)$ , and the functional value and the limiting value are the same, i.e.,

$$\lim_{(x_1, x_2, \dots, x_n) \rightarrow (a_1, a_2, \dots, a_n)} f(x_1, x_2, \dots, x_n) = f(a_1, a_2, \dots, a_n)$$

then we say  $f$  is continuous at the point  $(a_1, a_2, \dots, a_n)$ .

Consider the function  $f: D \rightarrow \mathbb{R}$  where  $D \subset \mathbb{R}^2$ , defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Answer the following question 8 to 13 using the above information.

If  $f$  approaches  $L$  as  $(x, y)$  approaches the origin along the  $X$ -axis, then find the value of  $L$ .

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

Along the  $X$ -axis,  $y = 0$ .

**Accepted Answers:**

(Type: Numeric) 0

1 point

If  $f$  approaches  $L$  as  $(x, y)$  approaches to the origin along the  $Y$ -axis, then find the value of  $L$ .

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

Along the  $Y$ -axis,  $x = 0$ .

**Accepted Answers:**

(Type: Numeric) 0

1 point

If  $f$  approaches  $L$  as  $(x, y)$  approaches to the origin along the straight line  $y = x$ , then find the value of  $L$ .

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

(Type: Numeric) 0.5

1 point

1 point

If  $f$  approaches  $L$  as  $(x, y)$  approaches to the origin along the straight line  $y = 2x$ , then find the value of  $L$ .

☐

0.

☐ $\frac{1}{2}$ .☐ $\frac{2}{9}$ .☐

11.

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

 $\frac{2}{9}$ .

1 point

If  $f$  approaches  $L$  as  $(x, y)$  approaches to the origin along the straight line  $y = mx$ , then find the value of  $L$ .

☐

0.

☐ $\frac{1}{m^2}$ .☐ $\frac{m}{1+m^3}$ .☐

11.

**No, the answer is incorrect.**

**Score: 0**

**Feedback:****Accepted Answers:**

$$\frac{m}{1+m^3}1+m3m.$$

1 point

Choose the set of correct options.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$  does not exist.
☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$ 
☐

f is not continuous at the origin.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{1}{2}$ 
**No, the answer is incorrect.****Score: 0****Accepted Answers:**
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$  does not exist.

f is not continuous at the origin.

1 point

Consider the function  $f: D \rightarrow \mathbb{R}$  where  $D \subset \mathbb{R}^2$ , defined as:

$$f(x,y) = \begin{cases} \frac{y}{x^2+y} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Choose the set of correct options.

[Hint: Approach the origin using the curves  $y = x^2$  and  $y = 2x^2$ .]☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$  does not exist.
☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$ 
☐

f is not continuous at the origin.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$ 
**No, the answer is incorrect.****Score: 0****Accepted Answers:**
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$  does not exist.

f is not continuous at the origin.

## Key Points: 6Key Points: 6

- Gradient of a function: Gradient of a function: If  $f$  is a scalar valued function defined on a domain  $D$  in  $R^n$ , then the gradient  $\nabla f$  is defined as the vector valued function  $(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n})$ .
- If  $\nabla f$  exists and is continuous in an open ball around the point  $(a_1, a_2, \dots, a_n)$ , then the directional derivative of the function  $f$  at  $(a_1, a_2, \dots, a_n)$  in the direction of a vector  $\mathbf{v}$  is given by the dot product of the gradient of the function at  $(a_1, a_2, \dots, a_n)$  with the unit vector in the direction of  $\mathbf{v}$  i.e.,

$$D_{\mathbf{v}} f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n) \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad D_{\mathbf{v}} f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n) \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

1 point

Find the directional derivative of the function  $f: R^2 \rightarrow R$  such that  $f(x, y) = 3x^2 - 2xy$  in the direction of the vector  $(3, 4)$ .

☐

$$2x - \frac{6y}{5} 2x - 56y.$$

☐

$$10x - 6y 10x - 6y.$$

☐

$$10x 10x.$$

☐

$$2x 2x.$$

☐

$$-\frac{6y}{5} - 56y.$$

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

$$2x - \frac{6y}{5} 2x - 56y.$$

1 point

Find the directional derivative of the function  $f: R^2 \rightarrow R$  such that  $f(x, y) = 3x^2 + 2y^2$  at the point  $(1, 1)$  along the direction of its gradient vector  $\nabla f(1, 1)$ .

[Hint: First find out the unit vector in the direction of the vector  $\nabla f(1, 1)$ .]

☐

$$22$$

☐

$$\sqrt{13} \quad 13$$

☐

$$2\sqrt{13} \quad 13$$

☐

$$2\sqrt{2} \quad 2$$

☐

$$\sqrt{2} \quad 2$$

**No, the answer is incorrect.****Score: 0**

### Accepted Answers:

$2\sqrt{132}$  13

$\sqrt{\phantom{x}}$